



# HUMBER

MENG 3050-Power Transmission Components

|                                     |                        |           |
|-------------------------------------|------------------------|-----------|
| Assignment 01:                      | Gear clutch            | Value: %5 |
| Submission Date: September 29, 2026 | Group: N/A             |           |
| Name<br>Michaël McCorkell           | Students:<br>N01500049 |           |

## LATE SUBMISSION:

|                  |                    |                       |                      |                      |
|------------------|--------------------|-----------------------|----------------------|----------------------|
| One day (-1mark) | Two days (-2marks) | Three days (-3 marks) | Four days (-4 marks) | Five days (-5 marks) |
|                  |                    |                       |                      |                      |

## MARKING Chart:

| Topics                                                  | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Worth | marking |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------|
| Using ANSI standard                                     | <ul style="list-style-type: none"><li>Inches</li><li>Correct scale</li><li>Correct file name and student name</li></ul>                                                                                                                                                                                                                                                                                                                                                                                      | 0.5   |         |
| File Managements                                        | <ul style="list-style-type: none"><li>Drawing to be saved as SolidWorks</li><li>Save your 2D drawings in PDF format and submit.</li><li>Wrong format cause us to lose all information sent.</li></ul>                                                                                                                                                                                                                                                                                                        | 0.5   |         |
| Only <b>Gear housing</b> detailed drawings are required |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |         |
| Housing Base<br>Part detailed drawing                   | <ul style="list-style-type: none"><li>Include complete part dimensions required for manufacturing.</li><li>Use dual dimensions where required.</li><li>Include appropriate thread and hole specifications.</li><li>Use the correct third-angle projection.</li><li>Include all required centerlines and center marks.</li><li>Include all necessary notes and comments and BOM.</li><li>Include all required fits and tolerances.</li></ul> <p>Note: Each missing item or error will be worth 0.25 marks</p> | 1.25  |         |
| Housing Lid<br>Part detailed drawing                    | <ul style="list-style-type: none"><li>Include complete part dimensions required for manufacturing.</li><li>Use dual dimensions where required.</li><li>Include appropriate thread and hole specifications.</li></ul>                                                                                                                                                                                                                                                                                         | 1.25  |         |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                        |     |  |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|
|                     | <ul style="list-style-type: none"> <li>• Use the correct third-angle projection.</li> <li>• Include all required centerlines and center marks.</li> <li>• Include all necessary notes and comments and BOM.</li> <li>• Include all required fits and tolerances.</li> </ul> <p>Note: Each missing item or error will be worth 0.25 marks</p>                                                                           |     |  |
| Notes and BOM       | <ul style="list-style-type: none"> <li>• Add necessary notes,</li> <li>• BOM has to include the following information: <ul style="list-style-type: none"> <li>○ Item number</li> <li>○ Part name</li> <li>○ material</li> <li>○ Size, part number or drawing number</li> </ul> </li> <li>• Quantity</li> <li>• Heat treatment</li> <li>• Surface finish</li> </ul> <p>(each mistake/ missing item worth 0.25 mark)</p> | 0.5 |  |
| Assembly 2D drawing | <ul style="list-style-type: none"> <li>• Correct assembly</li> <li>• All parts must be shown clearly,</li> <li>• Special Notes of assembly if required,</li> <li>• Dimensioning</li> </ul> <p>(each mistake/ missing item worth 0.25 mark)</p>                                                                                                                                                                         | 1.0 |  |

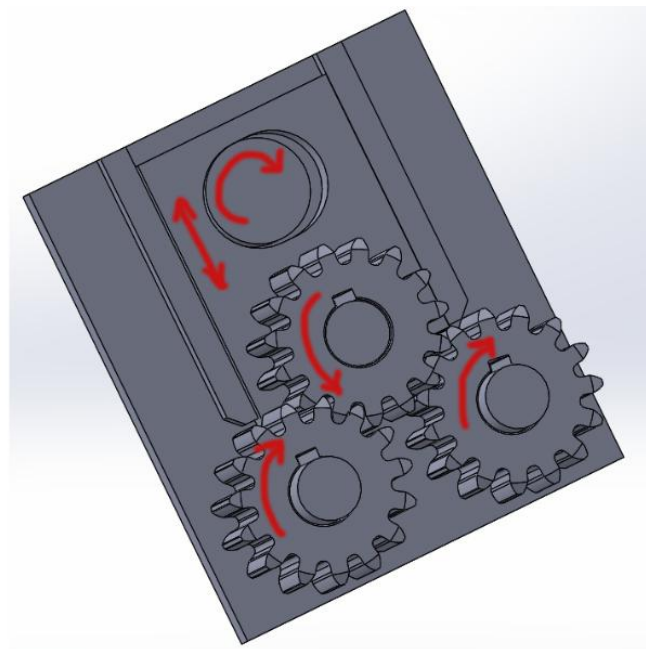
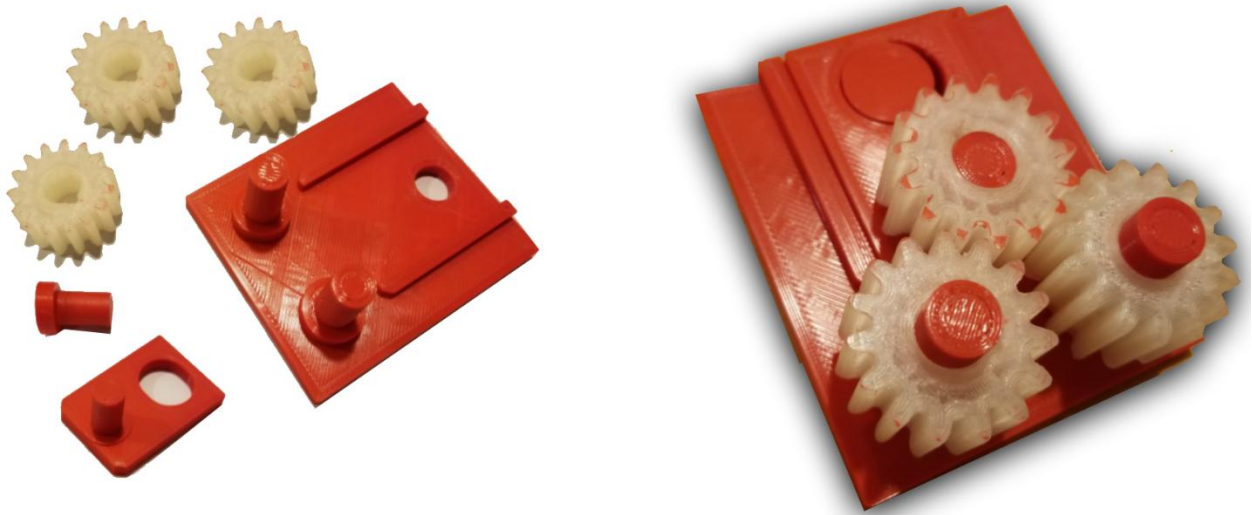
### HOSE REELS GEAR CLUTCH:

Hose reels are devices used in various industrial settings to efficiently store, dispense, and manage hoses and heavy cables. These reels are designed to accumulate the hose by winding it onto a motor-driven reel drum. The hose is unwound by manually pulling it from the reel.

For the unwinding operation, a gear clutch is required to disengage the motor from the reel drum, allowing the drum to rotate freely and making it easier for the operator to pull and unwind the hose.



During the design stage, prototypes were developed to validate the concept of using a cam mechanism to disengage the gears and facilitate the hose unwinding operation. As shown below:



## LEARNING OUTCOMES

- Design the missing components in 3D, the gear clutch housing, and create fully detailed 2D engineering drawings.
- Review and apply the theory of fit and tolerancing learned in class to the design and manufacturing of the components.
- Identify and use standard components from the SolidWorks Toolbox/library and third-party suppliers where appropriate.

## OBJECTIVE AND SUBMISSION

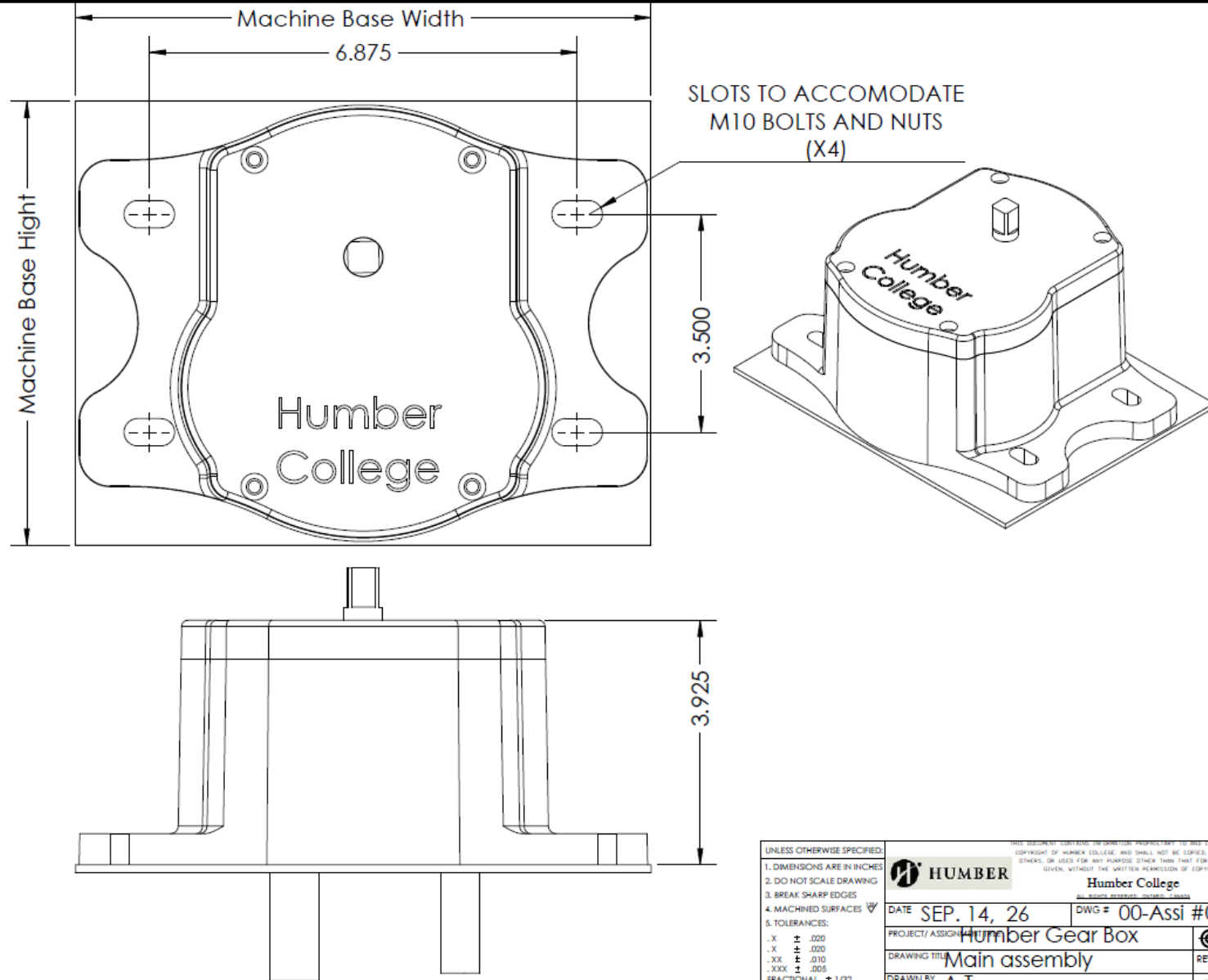
Use the STEP file attached to this assignment, then:

- Create a 2D drawing showing the correct geometric layout and Humber Template,
- Create a Bill of Materials (BOM) of the missing parts
- Whenever needed, use appropriate section views and detailed views to clearly show the part details.
- Submit your work as **PDF sheets** for the missing parts, including the main assembly and the **gear clutch housing** drawings.

## Requirements

Design the gearbox housing, considering the manufacturing, assembly, and overall functionality.

- Design the missing **gear clutch housing**, including both the **base and lid**. The housing must provide sufficient space and appropriate mounting features for all gearbox components, accommodate the **cam plate and slider mechanism**, and be designed to properly mount to the **main machine plate**.
- Identify all required fasteners, include them in the assembly drawing, and list them in the BOM.
- Produce a complete set of engineering drawings for manufacturing, including:
  - Main assembly drawing.
  - Bill of Materials (BOM), listing all manufactured and purchased parts.
  - Detailed manufacturing drawings for the gearbox housing part(s) only.
  - Apply fit and tolerance wherever applicable.



UNLESS OTHERWISE SPECIFIED:

1. DIMENSIONS ARE IN INCHES
  2. DO NOT SCALE DRAWING
  3. BREAK SHARP EDGES
  4. MACHINED SURFACES
  5. TOLERANCES:
- .X ± .000  
.X ± .000  
.XX ± .010  
.XXX ± .005  
FRACTIONAL ± 1/32  
ANGULAR ± 0.15°

|                                            |                 |
|--------------------------------------------|-----------------|
| <p><b>HUMBER</b></p> <p>Humber College</p> |                 |
| DATE                                       | SEP. 14, 26     |
| DWG #                                      | 00-Assi #01     |
| PROJECT / ASSIGNMENT                       | Humber Gear Box |
| DRAWING TITLE                              | Main assembly   |
| DRAWN BY                                   | A.T.            |
| REV                                        | 0               |
| SCALE                                      | 1:2             |